EX 10 – Demonstration of predictive analytics to analysis micro array data

AIM :

To demonstrate predictive analytics for analyzing microarray data to classify samples and identify important genes.

ALGORITHM :

1. Import libraries
2. Load microarray dataset
3. Handle missing values
4. Normalize data
5. Perform feature selection (select important genes)
6. Split dataset into training and testing sets
7. Train predictive model (e.g., SVM, Decision Tree, or KNN)
8. Predict class labels
9. Evaluate model (accuracy, confusion matrix)
10. Interpret results and identify significant patterns

CODE :

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

df = pd.read_csv("StudentsPerformance.csv")

df["high_performance"] = ((df["math score"] + df["reading score"] + df["writing score"]) /
3 > 70).astype(int)
```

```
X = df[["math score","reading score","writing score"]]
y = df["high_performance"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = RandomForestClassifier()
model.fit(X_train, y_train)

pred = model.predict(X_test)
print(accuracy_score(y_test, pred))
```

OUTPUT :

0.985

RESULT :

Thus the above program is executed and the result is obtained.